MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

THIRD SEMESTER EXAMINATION

(2024 ADMISSION ONWARDS)

MG3DSECFS200-TCP/IP AND NETWORK SECURITY

Duration: 2 hrs MaximumMarks:70

Students should attempt at least one question from each course outcome to enhance their overall outcome attain ability.

Part A

Short Answer Type Questions

Answer **any 10** Questions

Each question carries **2** mark

1. Explain how a LAN differs from a WAN.[Understand][1]

2. Compare OSI and TCP/IP models in terms of number of layers and purpose[Understand][1]

3. Classify the following networks as LAN, MAN, or WAN: a city-wide cable network, a college computer lab, and the internet. [Understand][1]

4. Identify various access control models used in security [Understand][2]

5. If a system detects an unusual login at midnight, what should it do? [Apply][2]

6. Classify firewalls. [Apply][2]

7. A system uses the same key for both encryption and decryption. Identify the type of encryption used and give one example. [Apply][3]

8. Compare public key cryptography with symmetric encryption [Apply][3]

9. Is Kerberos a secure method for authentication in modern networks? Support your answer [Apply][3]

10.Compare HTTP and HTTPS [Analyze][4]

11. Differentiate between Tunnel mode and Transport mode in IPsec.[Understand][4]

12. Identify the role of PGP in e-mail security.[Understand] [4]

[10x2=20]

Part B

Short Essay Type Questions

Answer **any 6** Questions

Each question carries **5** marks

13. Explain the layers of the TCP/IP model and describe the function of each layer. [Understand][1]

14. Break down the process of sending a message from application layer to the network layer in terms of data encapsulation. [Analyze][1]

15. Discuss the difference between passive and active attacks with examples. [Understand][2]

16. Indicate the importance of Password Management in computer security. [Understand][2]

17. Distinguish principles of symmetric encryption to design a secure message transmission system between two users. [Analyze][3]

18. Explain the structure of an X.509 certificate. [Understand][3]

19. How would you use SSH to connect safely to a remote computer? Explain the steps you would take.[Analyse][4]

20. Evaluate the effectiveness of S/MIME in securing email communication. Do you think it is suitable for widespread use.[Evaluate][4]

[6x5=30]

Part C

Essay Type Questions

Answer **any 2** Questions

Each question carries **10** marks

19. Explain the seven layers of the OSI reference model with the function of each layer [Understand][1]

20. Examine the role of Access Control Models for network security.[Analyze][2]

21. Evaluate the effectiveness of Public-Key Infrastructure (PKI) in securing internet communications. [Evaluate][4]

[2x10=20]